

Introduction to Robotics, Lecture 3

Differential Kinematics

Joint and tool velocities

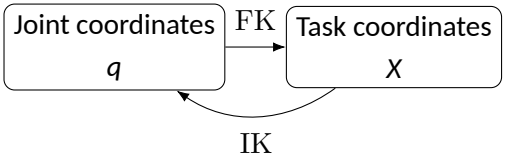
Jacobians and inverse differential maps

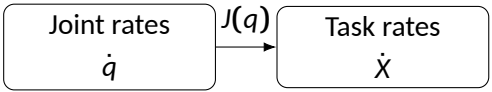
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Where This Lecture Starts

Pose maps are nonlinear; the velocity map is linear at a frozen configuration





$$\dot{x} = J(q)\dot{q}$$

At a fixed q, the map is linear in $\dot{q}$; moving q changes the map itself.

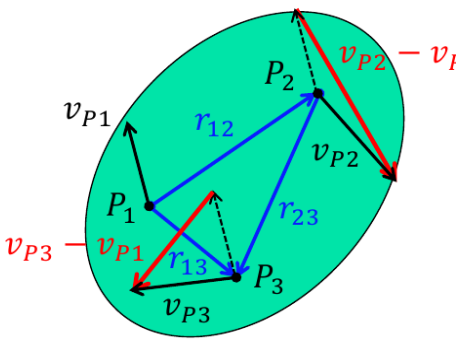
What Differential Kinematics Is

Instantaneous motion connects joint-space rates to linear and angular task motion

- Differential kinematics relates joint motion to end-effector linear and angular motion.
- The mapping is instantaneous: it is evaluated at one configuration, not over an entire path.
- Differentiate direct kinematics in time, or construct the map geometrically from joint axes.
- Rotation is the subtle part: coordinate rates and angular velocity are not generally the same quantity.

Angular Velocity of a Rigid Body

Rigidity makes one angular velocity serve every point of the same body



Every inter-point distance is fixed:

$$\|r_{ij}\| = \text{const} \quad \Rightarrow \quad v_{P_i} - v_{P_j} \perp r_{ij}.$$

Writing this condition for three points forces the pairwise angular rates to agree:

$$\omega_1 = \omega_2 = \omega.$$

A rigid body has one angular velocity, not one for every pair of points.

The Fundamental Kinematic Equation

Knowing one point velocity and the body angular velocity determines any other point

$$v_{P_j} = v_{P_i} + \omega \times r_{ij} = v_{P_i} + S(\omega) r_{ij}, \qquad \dot{r}_{ij} = \omega \times r_{ij}.$$

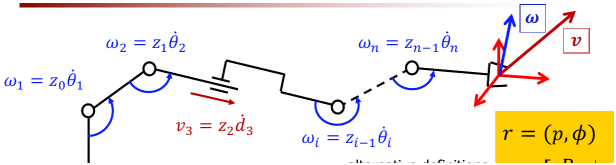
Two limiting motions

If $\omega = 0$, every point has the same velocity: pure translation. If two points are at rest, the body rotates about the line joining them.

The matrix $S(\omega)$ is skew-symmetric and represents cross product with ω.

One Joint, One Velocity Contribution

Revolute and prismatic joints contribute along the axis of their preceding frame



$$\omega_i = z_{i-1}\dot{\theta}_i \qquad v_i = z_{i-1}\dot{d}_i$$

A revolute joint contributes angular velocity; a prismatic joint contributes linear velocity. Contributions add because v and ω are vectors.

Why ω Is Not the Derivative of an Orientation

Angular velocity is a vector, whereas a minimal orientation triple is a coordinate choice

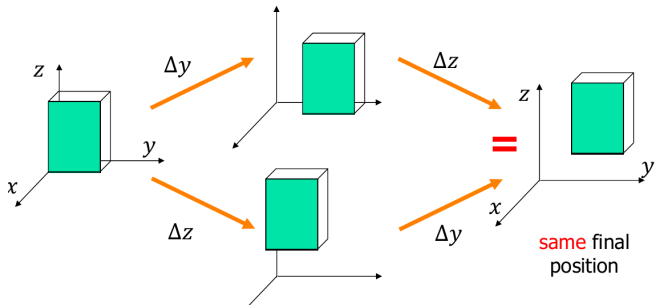
- Linear and angular velocities add in any order.
- A minimal orientation triple ϕ is not a vector: composing rotations is not addition of triples.
- Therefore, in general,

$$\omega \neq \dot{\phi}.$$

- Both (p, ϕ) and a rotation matrix R are valid orientation descriptions; their rates need different treatment.

Translations Always Commute

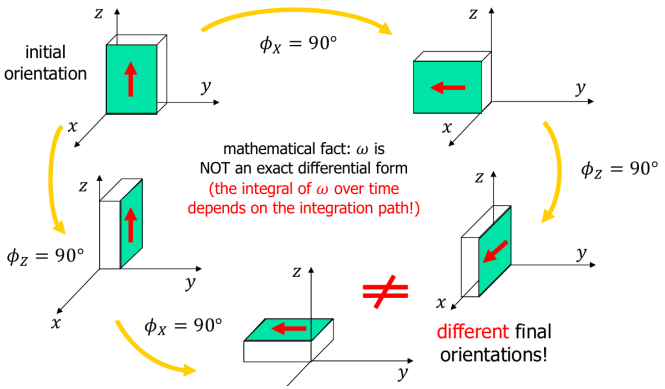
The order of two displacements does not affect the final position



Δy then $\Delta z = \Delta z$ then Δy. Finite and infinitesimal displacements both add as vectors; no ordering bookkeeping is needed.

Finite Rotations Do Not Commute

Changing the order of rotations about different axes changes the final orientation



$$R_X(\pi/2)R_Z(\pi/2) \neq R_Z(\pi/2)R_X(\pi/2).$$

Rotations about one fixed axis are the special commuting case.

ω Is Not an Exact Differential

Equal axis-wise areas of angular-rate profiles can lead to different orientations

Two rate profiles can satisfy

$$\int_0^T \omega \, dt$$

with the same vector result while producing different final rotation matrices R_f. The integral depends on the path through orientation space.

Coordinate rates are different:

$$\int_0^T \dot{\phi} \, dt = \phi_f - \phi_i.$$

This telescopes because $\dot{\phi}$ is an exact differential of its chosen coordinates.

Infinitesimal Rotations Do Commute

Keeping only first-order terms recovers an additive differential description

$$R(d\phi) = I + S(d\phi), \qquad S(d\phi) = \begin{bmatrix} 0 & -d\phi_Z & d\phi_Y \\ d\phi_Z & 0 & -d\phi_X \\ -d\phi_Y & d\phi_X & 0 \end{bmatrix}.$$

Multiplying elementary infinitesimal rotations in any order gives the same first-order matrix. Products of two small increments are discarded, not assumed to be zero.

Time Derivative of a Rotation Matrix

A matrix derivative reveals angular velocity without choosing orientation angles

Orthogonality gives a skew-symmetric product:

$$\dot{R}R^T = S, \qquad S = S(\omega).$$

Hence

$$\dot{R} = S(\omega)R \qquad \Longleftrightarrow \qquad S(\omega) = \dot{R}R^T.$$

This is the matrix form of angular velocity and will later extract the angular part of a Jacobian column.

Worked Example: Differentiating an Elementary Rotation

For a rotation about **x**, *the matrix identity returns the expected axis velocity*

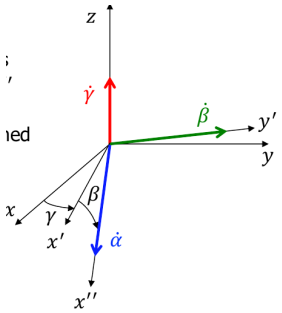
$$R_X(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \quad \Rightarrow \quad \dot{R}_X R_X^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\dot{\phi} \\ 0 & \dot{\phi} & 0 \end{bmatrix}.$$

$$\omega = \dot{\phi} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

More generally, an axis–angle rotation $R(r, \theta)$ has $\omega = \dot{\theta} r$ when its axis is fixed.

RPY Rates Against Angular Velocity

A coordinate singularity, not a robot singularity



The Z, new-Y, and newer-X rotation axes form the columns of T_{RPY}:

$$\omega = T_{\mathrm{RPY}}(\beta, \gamma) \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix}.$$

Its determinant is $\det T_{\mathrm{RPY}} = \cos\beta$. At $\beta = \pm\pi/2$, this representation cannot express every angular velocity by angle rates.

Two Jacobians, Not One

Analytic and geometric Jacobians answer different instantaneous questions

Analytic Jacobian

For task coordinates $r = f_r(q)$,

$$\dot{r} = \frac{\partial f_r}{\partial q}\dot{q} = J_r(q)\dot{q}.$$

It is obtained by differentiating the task map.

Geometric Jacobian

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} J_L(q) \\ J_A(q) \end{bmatrix} \dot{q} = J(q)\dot{q}.$$

It maps to the physical end-effector twist.

Their lower halves differ because $\omega \neq \dot{\phi}$ in general.

The Jacobian Equation

Each entry is a local sensitivity, and the matrix changes with configuration

$$\dot{X}_{6\times 1} = J_{6\times n}(q)\dot{q}_{n\times 1}, \qquad J = \left[\frac{\partial f_i}{\partial q_j}\right]_{\substack{i=1,\ldots,6\\ j=1,\ldots,n}}.$$

- A six-quantity task and a seven-joint robot give a 6×7 Jacobian.
- $J(q)$ is recomputed at every configuration; it is not a constant robot property.
- Column j records the effect of moving only joint j at that configuration.

Which Vector Is on the Left?

Coordinate rates use the analytic Jacobian; physical twist uses the geometric Jacobian

Coordinate-rate output

$$\begin{bmatrix} \dot{p} \\ \dot{\phi} \end{bmatrix} = J_r(q)\dot{q}.$$

The partial-derivative array is correct for the stack $(\dot{p}, \dot{\phi})$.

Physical-twist output

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}.$$

Here ω is not the derivative of three orientation coordinates.

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \mathrm{blkdiag}(I, T(\phi)) \begin{bmatrix} \dot{p} \\ \dot{\phi} \end{bmatrix}.$$

A Jacobian Is a Linear Map Between Two Spaces

The same configuration-dependent map appears in velocity, statics, and stiffness models

Motion

Small joint displacements map to small Cartesian displacements.

Statics

$$\tau = J^T(q)F.$$

Forces and moments map back to joint torques.

Stiffness

Joint and Cartesian stiffness are related locally through the same geometry.

The underlying kinematics is nonlinear, but its local linear map is enough for small changes at one q.

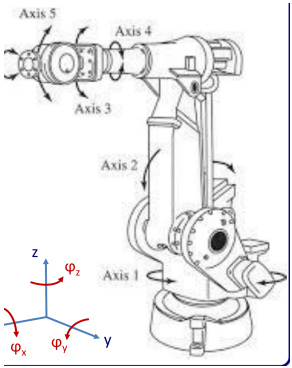
Physical Meaning of One Column

A column asks how the pose changes when only one joint moves

For the vertical base axis in the illustrated arm,

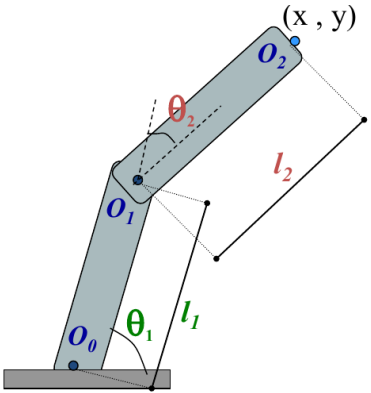
$$J_1 = \begin{bmatrix} -(L_3 + L_6)\sin q_1 \\ (L_3 + L_6)\cos q_1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

It sweeps the end effector in x and y, gives no vertical motion or tool tilt, and turns the tool about z.



Classical Approach: Differentiate the Direct Kinematics

For the planar 2R arm, the task coordinates are explicit functions of two angles



$$x = l_1 c_1 + l_2 c_{12}, \qquad y = l_1 s_1 + l_2 s_{12},$$

$$\varphi = \theta_1 + \theta_2, \qquad J = \frac{\partial(x, y, \varphi)}{\partial(\theta_1, \theta_2)}.$$

Differentiating column one means following both links; differentiating column two follows the second link only.

The 2R Analytic Jacobian, Evaluated

The chain rule makes the first column contain the contribution of both links

$$J = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \\ 1 & 1 \end{bmatrix}.$$

Read the columns physically

Changing θ_1 moves both links, whereas changing θ_2 moves only link 2. The last row is all ones because the tool angle is the sum of the two joint angles.

Geometric Rule: Columns from Joint Axes

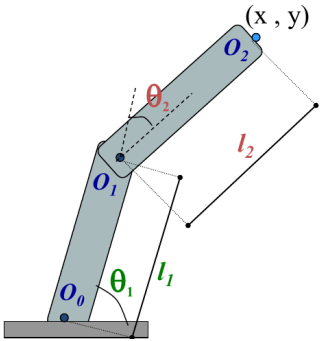
A column can be built directly from the relevant axis and the tool moment arm

$$J_i = \begin{cases} \begin{bmatrix} z_{i-1} \times (o_n - o_{i-1}) \\ z_{i-1} \end{bmatrix}, & \text{revolute joint,} \\ \begin{bmatrix} z_{i-1} \\ 0 \end{bmatrix}, & \text{prismatic joint.} \end{cases}$$

Here z_{i-1} is the third column and o_{i-1} the translation column of ${}^{0}T_{i-1}$. A prismatic joint has no angular contribution; a revolute joint has both a moment-arm velocity and an angular-axis velocity.

Setting Up the 2R Arm for the Geometric Rule

Planar joint axes and origins supply the moment arms



$$o_0 = 0, \quad o_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix},$$

$$o_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}.$$

$$z_0 = z_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Column One by Cross Product

Crossing the vertical base axis with the tool position rotates the planar moment arm

$$z_0 \times (o_2 - o_0) = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} \\ 0 \end{bmatrix}.$$

$$J_1 = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

The lower half is the joint axis itself, so this planar joint contributes only ω_z.

Column Two, and the Two Routes Agree

The elbow moment arm contains only link 2, giving the second column directly

$$J_2 = \begin{bmatrix} -a_2 s_{12} \\ a_2 c_{12} \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Its moment arm begins at the elbow origin, so link 1 does not appear.

Deleting the spatial rows this arm cannot move leaves

$$\begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \\ 1 & 1 \end{bmatrix}.$$

It agrees entry for entry with the differentiated result.

Differentiating the Chain Instead of the Equations

Replace only the elementary factor that contains the selected joint variable

$$T = T_{\text{base}}\,[\,T_1(q)\,H(q_i)\,T_2(q)\,]\,T_{\text{tool}}.$$
$$T'_{q_i} = T_{\text{base}}\,[\,T_1(q)\,H'(q_i)\,T_2(q)\,]\,T_{\text{tool}}.$$

Every other factor is untouched because it does not depend on q_i. One such product per joint gives all derivative matrices; this is the chain rule arranged for an implementation loop.

Reading a Jacobian Column out of the Derivative Matrix

Translation comes from the last column; angular velocity comes from a skew matrix

$$[J]_{1..3,i} = [T'_{q_i}]_{1..3,4}.$$

To get the angular half, remove the current orientation:

$$S(\omega) = T'_{q_i} T^{-1}$$

in the upper-left 3×3 block. Read the vector as

$$\omega_x = S_{32}, \qquad \omega_y = S_{13}, \qquad \omega_z = S_{21}.$$

Elementary Translations and Their Derivatives

A prismatic elementary transform differentiates to one fixed matrix entry

$$\frac{\partial T_x}{\partial x} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial T_y}{\partial y} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial T_z}{\partial z} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The derivative is independent of the translation value: it simply selects the corresponding coordinate in the last column.

Elementary Rotations and Their Derivatives

At zero angle, rotation derivatives are the embedded skew generators

$$R_x'(0) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad R_y'(0) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad R_z'(0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

In homogeneous-transform form, add a zero fourth row and column. These are exactly the skew generators from the infinitesimal-rotation construction; zero angle is a special case, not the general derivative.

Three Routes, One Jacobian

Choose the construction that fits the available model and the task at hand

Task equations	Joint axes	Transform chain
Differentiate closed-form kinematics. Shortest on paper, but requires explicit task equations.	Build columns from axes and moment arms. No symbolic differentiation is needed.	Replace an elementary factor by its derivative. Mechanical and suitable for code.

The first route gives an analytic Jacobian; the geometric and transform-chain routes give the geometric Jacobian. At the same configuration they are related by the representation-rate map.

The Inverse Differential Problem

Given an end-effector velocity, find joint rates that realize it

$$v = J(q)\dot{q} \qquad \Rightarrow \qquad \dot{q} = J^{-1}(q)v$$

only when J is square and nonsingular. Two failures are fundamental:

- near a singularity, an inverse may yield physically excessive joint rates;
- for a redundant robot, J is rectangular and has no ordinary inverse.

Inversion as a Minimum-Norm Problem

The pseudoinverse chooses the smallest joint-rate vector among the available solutions

$$\min \frac{1}{2}\|\dot{q}\|^2 \qquad \text{subject to} \qquad J\dot{q} = v.$$

Its solution is

$$\dot{q} = J^{\#} v.$$

If the demand is infeasible, the same construction first minimizes the task-space error and then chooses the smallest-norm joint vector among those minimizers.

The Pseudoinverse

One unique matrix always exists; convenient product formulas require full rank

$$JJ^{\#}J = J, \quad J^{\#}JJ^{\#} = J^{\#}, \quad (JJ^{\#})^T = JJ^{\#}, \quad (J^{\#}J)^T = J^{\#}J.$$

Square, full rank	Full row rank	Full column rank
$J^{\#} = J^{-1}$	$J^{\#} = J^T(JJ^T)^{-1}$	$J^{\#} = (J^TJ)^{-1}J^T$

At rank deficiency, compute it from a singular-value decomposition rather than these products.

A Singular 2R Arm, Worked

At full extension the two Jacobian columns are parallel and the rank is one

$$J = \begin{bmatrix} -2s_1 & -s_1 \\ 2c_1 & c_1 \end{bmatrix}, \qquad J^{\#} = \frac{1}{5}\begin{bmatrix} -2s_1 & 2c_1 \\ -s_1 & c_1 \end{bmatrix}.$$

Neither full-rank product formula applies. Instead, $JJ^{\#}$ and $J^{\#}J$ are symmetric and the four Penrose relations hold.

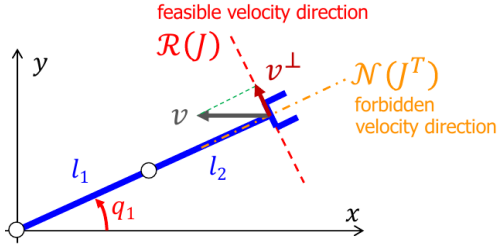
Feasible and Forbidden Directions

At a singular configuration, the pseudoinverse projects a requested velocity onto the range of J

$$J\dot{q} = JJ^{\#}v = v^{\perp}.$$

The range of J contains feasible task velocities; its orthogonal complement contains directions that no joint rate can produce.

$$q_1 = \pi/6: \quad \dot{q} = \begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix}, \quad v^{\perp} = \begin{bmatrix} -1/8 \\ \sqrt{3}/8 \end{bmatrix}.$$



Inverting at Higher Differential Order

Differentiating the velocity relation brings in the Jacobian rate and acceleration

$$\ddot{q} = J_r^{-1}(q)\left(\ddot{r} - \dot{J}_r(q)\dot{q}\right).$$

One more product rule gives

$$\dddot{q} = J_r^{-1}(q)\left(\dddot{r} - 2\dot{J}_r(q)\ddot{q} - \ddot{J}_r(q)\dot{q}\right).$$

The inverse, or a pseudoinverse when necessary, remains the leading map at each differential order. Integrating a higher-order command down produces smoother joint motion.

What This Leaves Open

The Jacobian exposes the next control and modelling questions

- How can singular configurations be found and avoided?
- What should a redundant arm do with its unused joint-rate freedom?
- How does the statics duality map forces to joint torques?
- How should numerical conditioning near singularities be handled, including damped alternatives?

Reference Materials

- A. Klimchik — L3 Differential kinematics v01
- A. De Luca — 11 DifferentialKinematics
- A. De Luca — 12a InverseDifferentialKinematics

Deserve “A” grade!

– Oleg Bulichev

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